Lesson 1 · What is a sensor? · for your pupils

Human sensors and real sensors

40 minutes · Ages 10–16 · adapt for other ages below · No electricity needed · No devices needed · Photocopy-friendly

Big question

Why do we need sensors when we have our own senses?

Pupils will be able to...

- Explain that a sensor notices something in the world and turns it into a number.
- Name four sensors found at school, at home or in a phone.
- Sort everyday objects into sensors, displays and actuators.

You need

- Three bowls or buckets of water: cold, room temperature and warm (not hot)
- A thermometer, if the school has one
- 12 picture cards per group: sensors, displays and actuators (see prep)
- Board and chalk

Before class

Fill the bowls just before class; the warm water should feel comfortable, not hot. Draw 12 cards per group: thermometer, solar lamp sensor, soil probe, microphone, camera; TV screen, scale dial, scoreboard; pump, fan motor, buzzer, loudspeaker.

Time	What you do	What you can say	Pupils
Warm-up 0–5 min	Ask pupils to close their eyes for 30 seconds and notice everything they can: sounds, warmth, the chair, smells. Then list what they noticed and which body part noticed it.	*"Your body is full of sensors. What did your ears, skin and nose notice just now?"*	Name things they noticed and the body part that sensed each one.
Bowls test 5–15 min	A volunteer holds one hand in the cold bowl and one in the warm bowl for a minute, then puts both in the middle bowl. Repeat with two more volunteers.	*"Both hands are in the same water. So why does each hand tell a different story?"*	Predict, watch and describe what each hand feels.
Teach 15–23 min	Read the thermometer in the middle bowl, or explain it would give one number whoever held it. Write sensor, display and actuator on the board, each with a drawing and an example.	*"A sensor notices something in the world and turns it into a number. A display shows the number. An actuator acts."*	Copy the three words with a drawing and one example of each.
Card sort 23–35 min	Groups sort the 12 cards into three piles: sensors, displays and actuators. Walk around and ask each group to explain one tricky card.	*"Does this card notice something, show something or do something?"*	Sort the cards, agree as a group and explain their choices.
Wrap-up 35–40 min	Each group shares one card they argued about. Exit ticket: each pupil writes one sensor, what it senses and where they have seen it.	*"Tonight, look for one sensor at home. Tomorrow, tell us what it senses."*	Write the exit ticket; share one answer.

Check understanding
Listen for pupils saying what a sensor notices and that it gives a number — not just "it's electronic". If a group puts the pump with the sensors, ask: "Does it notice the water, or move it?"

Exit ticket
One sensor + what it senses + where I saw it.

Common mix-ups

Pupils may think…	Help them see…
Sensors always need electricity.	A glass thermometer has no battery: the liquid expands and climbs the scale. Many sensors use electricity, but not all.
A sensor and a display are the same thing.	They often sit in one device, but the jobs differ. A thermometer's bulb senses; its marked scale displays.
Our hands can measure temperature exactly.	The bowls test shows hands can be fooled. A thermometer gives the same number whoever reads it.

Adapt it

Younger pupils
Ages 6–9: do the bowls test, then play "sense it or do it?": hold up a card; pupils touch their eyes for a sensor and wave their arms for an actuator.

Older pupils and university
Pupils draw a device they know — a solar lamp, a phone, an automatic tap — and label each part as sensor, display or actuator. Discuss: where does the deciding happen?

Large classes
Run the bowls test once at the front with two volunteers. Read the cards aloud; pupils point up for sensor, forward for display and sideways for actuator.

No board or no space
Mark three spots in the yard — a stone, a stick, a leaf — for sensor, display and actuator. Call out an object; pupils walk to the right spot.

Short on time? Do the bowls test with one volunteer, define a sensor, and sort six cards as a whole class (20 minutes).

What you need to know

A sensor notices something in the world — heat, light, moisture, movement or sound — and turns it into a number or signal. Displays show that number; actuators act on it.

Sensor
A part that notices something in the world, such as heat, light or movement, and turns it into a number or signal.

Signal
A message carried by a change, such as a small electric current that grows stronger or weaker as the light changes.

Display
The part that shows a reading so people can see it, such as a screen, a dial or the scale on a thermometer.

Actuator
A part that does something in the world when it gets a signal, such as a pump, a motor or a buzzer.

My notes — how did it go?